Infrared Thermometer User Manual-485

1. Product description

Thank you for choosing an online infrared thermometer.

A series of infrared thermometers can calculate the surface temperature of an object by measuring the intensity of infrared radiation emitted by the target without touching the target. Non-contact temperature measurement is the biggest advantage of infrared thermometers, allowing users to easily measure targets that are difficult to approach or move.

The series of infrared thermometers are integrated integrated infrared temperature sensors. The sensor, optical system and electronic circuit are integrated in a stainless steel housing; it is easy to install, and the standard thread on the stainless steel housing can be quickly connected to the installation site; there are also various types Options (such as purging device, mounting bracket, three-dimensional adjustable mounting bracket, purge protection cover, etc.) to meet the requirements of various working conditions.

2. Physical description

a. Basic performance

Protection level IP65 (NEMA-4)
Ambient temperature 0 ~60°C
Storage temperature -20 ~ 80°C
Relative humidity 10 – 95% (No condensation)
Material stainless steel
Size M18×1 , Long 106 *18mm
Cable length 1.5 m (standard) Other lengths can be customized.

b. Electrical parameters

Working power supply 5-12VDC
Maximum current 50mA
Output 485

c. Measurement parameters

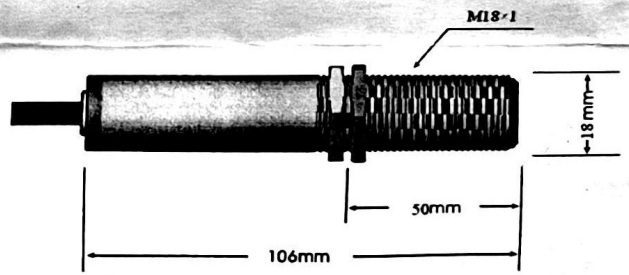
Spectral range 8 ~ 14 μm
Temperature range 0-500°C
Optical resolution 20:1
Response time 100 ms (95%)
Temperature measurement accuracy
±1% of the measured value or ±2℃, whichever is greater
Repeatability
±0.5% of measured value or ±2℃, whichever is greater
Launch rate 0.95 fixed

d. Wiring method

Signal	Line color	Function
Output 485	**Red line**	+ 5-12VDC Power positive
	black line	- 5-12VDC Power negative
	Yellow line	485T-/B
	Blue Line	485T+/A

3. Dimensions

Dimensions of infrared thermometer



4. Application field

Power distribution cabinets, electric power, tobacco, heating furnaces, food, rubber, paper, plastics, packaging, printing and dyeing textiles, electrical, composite materials, equipment supporting, metallurgy, petrochemical systems, various furnaces, heating furnaces, railways, and packaging machinery Temperature monitoring and so on.

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The general communication format of digital infrared temperature sensor is:

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x54	0x50	Sensor address	Command word	data	data	Checksum

4.1 Sensor address setting (**Sensor factory default address: 01**)

The 485 bus can connect up to 256 sensors online at the same time, but the address of each sensor must be different, otherwise it will cause confusion in data transmission)

Address change command format:

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
0x54	0x50	Sensor address	0xF0	new address	Checksum

4.2 Get the temperature measured by the sensor

Command format:

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5
0x54	0x50	Sensor address	0xF1	Checksum

Return data format:

Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7
0x54	0x50	Sensor address	0xF1	DataH	DataL	Checksum

Target temperature = (DataH*256+ DataL) / 10 (℃)

Checksum: Hexadecimal addition.

54+50+01+F1=196 The checksum is 96

Checksum: 54+50+ Sensor address +F1+datah+datal

Such as: 54+50+01+F1+00+72=208 The checksum is 08

default 9600 8 N 1 Baud rate: 9600 N: No check digit

8: 8 data bits 1: 1 stop bit

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